

Ejemplos del libro

Halliday, Resnick y Walker — *Fundamentals of Physics*

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Dos bloques que se empujan entre sí

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Sample Problem 5.07 Acceleration of block pushing on block

Some homework problems involve objects that move together, because they are either shoved together or tied together. Here is an example in which you apply Newton's second law to the composite of two blocks and then to the individual blocks.

In Fig. 5-18*a*, a constant horizontal force $\vec{F}_{\text{app}}$ of magnitude 20 N is applied to block *A* of mass $m_A = 4.0$ kg, which pushes against block *B* of mass $m_B = 6.0$ kg. The blocks slide over a frictionless surface, along an *x* axis.

(a) What is the acceleration of the blocks?

Serious Error: Because force $\vec{F}_{\text{app}}$ is applied directly to block *A*, we use Newton's second law to relate that force to the acceleration $\vec{a}$ of block *A*. Because the motion is along the *x* axis, we use that law for *x* components ($F_{\text{net},x} = ma_x$), writing it as

$$F_{\text{app}} = m_A a.$$

However, this is seriously wrong because $\vec{F}_{\text{app}}$ is not the only horizontal force acting on block *A*. There is also the force $\vec{F}_{AB}$ from block *B* (Fig. 5-18*b*).

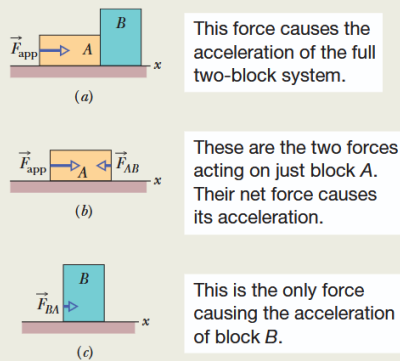


Figure 5-18 (a) A constant horizontal force $\vec{F}_{\text{app}}$ is applied to block *A*, which pushes against block *B*. (b) Two horizontal forces act on block *A*. (c) Only one horizontal force acts on block *B*.

Dead-End Solution: Let us now include force $\vec{F}_{AB}$ by writing, again for the *x* axis,

$$F_{\text{app}} - F_{AB} = m_A a.$$

(We use the minus sign to include the direction of $\vec{F}_{AB}$.) Because F_{AB} is a second unknown, we cannot solve this equation for *a*.

Successful Solution: Because of the direction in which force $\vec{F}_{\text{app}}$ is applied, the two blocks form a rigidly connected system. We can relate the net force *on the system* to the acceleration *of the system* with Newton's second law. Here, once again for the *x* axis, we can write that law as

$$F_{\text{app}} = (m_A + m_B) a,$$

where now we properly apply $\vec{F}_{\text{app}}$ to the system with total mass $m_A + m_B$. Solving for *a* and substituting known values, we find

$$a = \frac{F_{\text{app}}}{m_A + m_B} = \frac{20 \text{ N}}{4.0 \text{ kg} + 6.0 \text{ kg}} = 2.0 \text{ m/s}^2. \quad (\text{Answer})$$

Thus, the acceleration of the system and of each block is in the positive direction of the *x* axis and has the magnitude 2.0 m/s².

(b) What is the (horizontal) force $\vec{F}_{BA}$ on block *B* from block *A* (Fig. 5-18*c*)?

KEY IDEA

We can relate the net force on block *B* to the block's acceleration with Newton's second law.

Calculation: Here we can write that law, still for components along the *x* axis, as

$$F_{BA} = m_B a,$$

which, with known values, gives

$$F_{BA} = (6.0 \text{ kg})(2.0 \text{ m/s}^2) = 12 \text{ N}. \quad (\text{Answer})$$

Thus, force $\vec{F}_{BA}$ is in the positive direction of the *x* axis and has a magnitude of 12 N.

QUÉ MUESTRA

Que la segunda ley se aplica al *sistema* o a *un* cuerpo, pero no a una mezcla de los dos. El libro escribe primero el error ($F_{\text{app}} = m_A a$, como si *A* estuviera solo), después el callejón sin salida ($F_{\text{app}} - F_{AB} = m_A a$, que agrega una incógnita nueva), y recién entonces la salida buena: tomar los dos bloques como un solo cuerpo de masa $m_A + m_B$ para sacar *a*, y volver al bloque *B* aislado para sacar la fuerza de contacto.

LA CLAVE

Con el sistema entero, las fuerzas de contacto entre *A* y *B* son internas y se cancelan de a pares; por eso no aparecen. Reaparecen apenas aislás uno solo de los dos, y ahí $F_{BA} = m_B a$ sale de una.

SE PARECE A

P2 ej. 5 es el mismo problema con otros números, incluido el inciso que pide mostrar que aplicando *F* del otro lado la fuerza de contacto cambia — que es exactamente lo que este ejemplo deja listo para responder. También **P2 ej. 3** (bloque tirado por una sogá con masa: $F_{BC}/F = M/(M + m)$) es esta misma cuenta) y **P2 ej. 7** (tres bloques, T_1 y T_2).

¿Estática o cinética? La pregunta que va primero

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Sample Problem 6.01 Angled force applied to an initially stationary block

This sample problem involves a tilted applied force, which requires that we work with components to find a frictional force. The main challenge is to sort out all the components. Figure 6-3a shows a force of magnitude $F = 12.0\text{ N}$ applied to an 8.00 kg block at a downward angle of $\theta = 30.0^\circ$. The coefficient of static friction between block and floor is $\mu_s = 0.700$; the coefficient of kinetic friction is $\mu_k = 0.400$. Does the block begin to slide or does it remain stationary? What is the magnitude of the frictional force on the block?

KEY IDEAS

(1) When the object is stationary on a surface, the static frictional force balances the force component that is attempting to slide the object along the surface. (2) The maximum possible magnitude of that force is given by Eq. 6-1 ($f_{s,\max} = \mu_s F_N$). (3) If the component of the applied force along the surface exceeds this limit on the static friction, the block begins to slide. (4) If the object slides, the kinetic frictional force is given by Eq. 6-2 ($f_k = \mu_k F_N$).

Calculations: To see if the block slides (and thus to calculate the magnitude of the frictional force), we must compare the applied force component F_x with the maximum magnitude $f_{s,\max}$ that the static friction can have. From the triangle of components and full force shown in Fig. 6-3b, we see that

$$F_x = F \cos \theta \\ = (12.0\text{ N}) \cos 30^\circ = 10.39\text{ N}. \quad (6-3)$$

From Eq. 6-1, we know that $f_{s,\max} = \mu_s F_N$, but we need the magnitude F_N of the normal force to evaluate $f_{s,\max}$. Because the normal force is vertical, we need to write Newton's second law ($F_{\text{net},y} = ma_y$) for the vertical force components acting on the block, as displayed in Fig. 6-3c. The gravitational force with magnitude mg acts downward. The applied force has a downward component $F_y = F \sin \theta$. And the vertical acceleration a_y is just zero. Thus, we can write Newton's

second law as

$$F_N - mg - F \sin \theta = m(0), \quad (6-4)$$

which gives us

$$F_N = mg + F \sin \theta. \quad (6-5)$$

Now we can evaluate $f_{s,\max} = \mu_s F_N$:

$$f_{s,\max} = \mu_s (mg + F \sin \theta) \\ = (0.700)((8.00\text{ kg})(9.8\text{ m/s}^2) + (12.0\text{ N})(\sin 30^\circ)) \\ = 59.08\text{ N}. \quad (6-6)$$

Because the magnitude F_x ($= 10.39\text{ N}$) of the force component attempting to slide the block is less than $f_{s,\max}$ ($= 59.08\text{ N}$), the block remains stationary. That means that the magnitude f_s of the frictional force matches F_x . From Fig. 6-3d, we can write Newton's second law for x components as

$$F_x - f_s = m(0), \quad (6-7)$$

and thus $f_s = F_x = 10.39\text{ N} \approx 10.4\text{ N}$. (Answer)

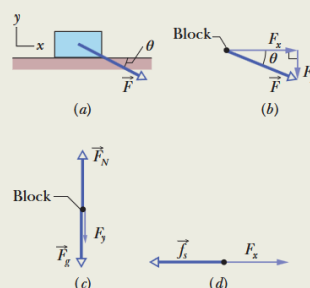


Figure 6-3 (a) A force is applied to an initially stationary block. (b) The components of the applied force. (c) The vertical force components. (d) The horizontal force components.

QUÉ MUESTRA

Que antes de escribir cualquier fuerza de rozamiento hay que decidir si el cuerpo desliza. Se compara la componente que empuja, $F_x = F \cos \theta = 10,39\text{ N}$, contra el máximo que la estática puede entregar, $f_{s,\max} = \mu_s F_N = 59,08\text{ N}$. Como $F_x < f_{s,\max}$ el bloque no arranca, y entonces el rozamiento vale $f_s = F_x = 10,4\text{ N}$ — no $\mu_s F_N$, que es sólo la cota.

LA CLAVE

La normal no es mg . La fuerza está aplicada hacia abajo, así que $F_N = mg + F \sin \theta = 84,4\text{ N}$, y no los $78,4\text{ N}$ del peso solo. Inclined hacia arriba sería $F_N = mg - F \sin \theta = 72,4\text{ N}$ y el umbral bajaría a $50,7\text{ N}$: el mismo módulo de fuerza, inclinado para el otro lado, corre el umbral casi 9 N . Acá no alcanza para cambiar la respuesta, pero es lo primero que hay que recalculer cuando el ángulo cambia de signo.

SE PARECE A

P2 ej. 8 (bloque contra la pared con $\mu_e = 0,6$ y $\mu_d = 0,4$, que pregunta literalmente «¿se comenzará a mover?») y **P2 ej. 19** (arrastrar el bloque B , con μ_e y μ_d dados por separado justamente para obligar a hacer esta comparación). **P2 ej. 9** es el caso espejado: F tira hacia arriba y hay que maximizar la aceleración en θ , donde todo el juego está en cómo $F \sin \theta$ le baja la normal.

Círculo vertical: la condición en el punto más alto

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Sample Problem 6.04 Vertical circular loop, Diavolo

Largely because of riding in cars, you are used to horizontal circular motion. Vertical circular motion would be a novelty. In this sample problem, such motion seems to defy the gravitational force.

In a 1901 circus performance, Allo “Dare Devil” Diavolo introduced the stunt of riding a bicycle in a loop-the-loop (Fig. 6-9a). Assuming that the loop is a circle with radius $R = 2.7$ m, what is the least speed v that Diavolo and his bicycle could have at the top of the loop to remain in contact with it there?

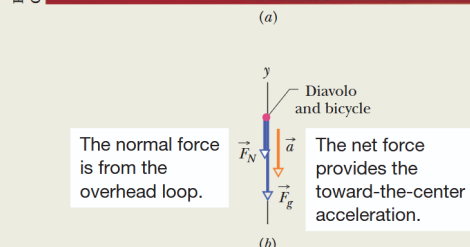
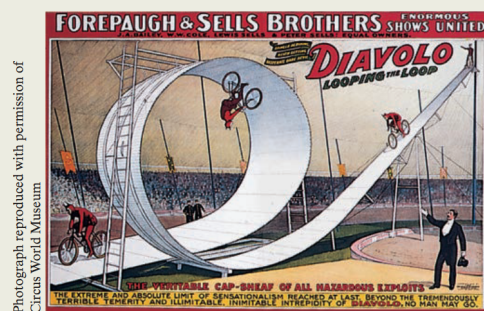


Figure 6-9 (a) Contemporary advertisement for Diavolo and (b) free-body diagram for the performer at the top of the loop.

KEY IDEA

We can assume that Diavolo and his bicycle travel through the top of the loop as a single particle in uniform circular motion. Thus, at the top, the acceleration $\vec{a}$ of this particle must have the magnitude $a = v^2/R$ given by Eq. 6-17 and be directed downward, toward the center of the circular loop.

Calculations: The forces on the particle when it is at the top of the loop are shown in the free-body diagram of Fig 6-9b. The gravitational force $\vec{F}_g$ is downward along a y axis; so is the normal force $\vec{F}_N$ on the particle from the loop (the loop can push down, not pull up); so also is the centripetal acceleration of the particle. Thus, Newton's second law for y components ($F_{\text{net},y} = ma_y$) gives us

$$-F_N - F_g = m(-a)$$

$$\text{and} \quad -F_N - mg = m\left(-\frac{v^2}{R}\right). \quad (6-19)$$

If the particle has the *least speed* v needed to remain in contact, then it is on the *verge of losing contact* with the loop (falling away from the loop), which means that $F_N = 0$ at the top of the loop (the particle and loop touch but without any normal force). Substituting 0 for F_N in Eq. 6-19, solving for v , and then substituting known values give us

$$v = \sqrt{gR} = \sqrt{(9.8 \text{ m/s}^2)(2.7 \text{ m})} = 5.1 \text{ m/s.} \quad (\text{Answer})$$

Comments: Diavolo made certain that his speed at the top of the loop was greater than 5.1 m/s so that he did not lose contact with the loop and fall away from it. Note that this speed requirement is independent of the mass of Diavolo and his bicycle. Had he feasted on, say, pierogies before his performance, he still would have had to exceed only 5.1 m/s to maintain contact as he passed through the top of the loop.

QUÉ MUESTRA

Cómo se traduce «no se despegas de la vía» a una ecuación. Arriba del rizo la normal y el peso apuntan las dos hacia el centro, así que $F_N + mg = m v^2/R$. La velocidad mínima es aquella en la que la vía ya no necesita empujar: $F_N = 0$, y queda $v = \sqrt{gR} = 5,1$ m/s.

LA CLAVE

La masa se cancela: la velocidad mínima no depende de cuánto pese el que da la vuelta. Y el signo importa — arriba las dos fuerzas suman, abajo se restan ($F_N - mg = m v^2/R$), que es por qué la normal (o la tensión) es máxima en el punto más bajo y mínima en el más alto.

SE PARECE A

P2 ej. 13 pide la tensión de la cuerda en un círculo vertical para un ángulo θ cualquiera: este ejemplo es el caso particular $\theta = 0$ (arriba), y sirve para chequear que la expresión general dé bien en ese extremo. En **P3 ej. 1** la tensión en el punto más bajo del péndulo es el otro extremo, y **P3 ej. 7** usa la misma condición para el carro de la montaña rusa.

OJO CON

El enunciado del libro dice «permanecer en contacto», no «dar la vuelta». Son la misma condición sólo porque la vía empuja y no tira. Con una barra rígida o un carro encarrilado, que sí pueden tirar hacia afuera, F_N puede ser negativa y la velocidad mínima baja a cero.

Curva plana: el rozamiento haciendo de centrípeta

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Sample Problem 6.05 Car in flat circular turn

Upside-down racing: A modern race car is designed so that the passing air pushes down on it, allowing the car to travel much faster through a flat turn in a Grand Prix without friction failing. This downward push is called *negative lift*. Can a race car have so much negative lift that it could be driven upside down on a long ceiling, as done fictionally by a sedan in the first *Men in Black* movie?

Figure 6-10a represents a Grand Prix race car of mass $m = 600$ kg as it travels on a flat track in a circular arc of radius $R = 100$ m. Because of the shape of the car and the wings on it, the passing air exerts a negative lift $\vec{F}_L$ downward on the car. The coefficient of static friction between the tires and the track is 0.75. (Assume that the forces on the four tires are identical.)

(a) If the car is on the verge of sliding out of the turn when its speed is 28.6 m/s, what is the magnitude of the negative lift $\vec{F}_L$ acting downward on the car?

KEY IDEAS

1. A centripetal force must act on the car because the car is moving around a circular arc; that force must be directed toward the center of curvature of the arc (here, that is horizontally).
2. The only horizontal force acting on the car is a frictional force on the tires from the road. So the required centripetal force is a frictional force.
3. Because the car is not sliding, the frictional force must be a *static* frictional force $\vec{f}_s$ (Fig. 6-10a).
4. Because the car is on the verge of sliding, the magnitude f_s is equal to the maximum value $f_{s,\max} = \mu_s F_N$, where F_N is the magnitude of the normal force $\vec{F}_N$ acting on the car from the track.

Radial calculations: The frictional force $\vec{f}_s$ is shown in the free-body diagram of Fig. 6-10b. It is in the negative direction of a radial axis r that always extends from the center of curvature through the car as the car moves. The force produces a centripetal acceleration of magnitude v^2/R . We can relate the force and acceleration by writing Newton's second law for components along the r axis ($F_{\text{net},r} = ma_r$) as

$$-f_s = m \left(-\frac{v^2}{R} \right). \quad (6-20)$$

Substituting $f_{s,\max} = \mu_s F_N$ for f_s leads us to

$$\mu_s F_N = m \left(\frac{v^2}{R} \right). \quad (6-21)$$

Vertical calculations: Next, let's consider the vertical forces on the car. The normal force $\vec{F}_N$ is directed up, in the positive direction of the y axis in Fig. 6-10b. The gravitational force $\vec{F}_g = m\vec{g}$ and the negative lift $\vec{F}_L$ are directed down. The acceleration of the car along the y axis is zero. Thus we can write Newton's second law for components along the y axis ($F_{\text{net},y} = ma_y$) as

$$F_N - mg - F_L = 0, \quad (6-22)$$

or

$$F_N = mg + F_L.$$

Combining results: Now we can combine our results along the two axes by substituting Eq. 6-22 for F_N in Eq. 6-21. Doing so and then solving for F_L lead to

$$\begin{aligned} F_L &= m \left(\frac{v^2}{\mu_s R} - g \right) \\ &= (600 \text{ kg}) \left(\frac{(28.6 \text{ m/s})^2}{(0.75)(100 \text{ m})} - 9.8 \text{ m/s}^2 \right) \\ &= 663.7 \text{ N} \approx 660 \text{ N}. \end{aligned} \quad (\text{Answer})$$

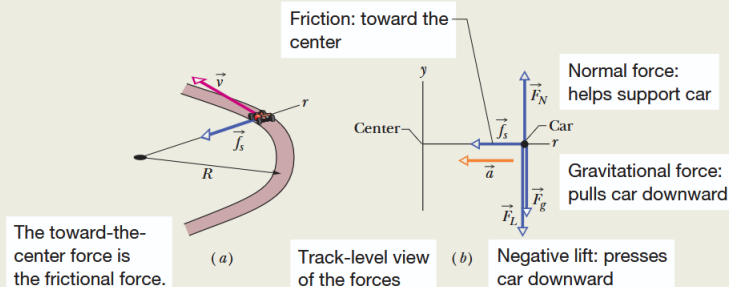


Figure 6-10 (a) A race car moves around a flat curved track at constant speed v . The frictional force $\vec{f}_s$ provides the necessary centripetal force along a radial axis r . (b) A free-body diagram (not to scale) for the car, in the vertical plane containing r .

(b) The magnitude F_L of the negative lift on a car depends on the square of the car's speed v^2 , just as the drag force does (Eq. 6-14). Thus, the negative lift on the car here is greater when the car travels faster, as it does on a straight section of track. What is the magnitude of the negative lift for a speed of 90 m/s?

KEY IDEA

F_L is proportional to v^2 .

Calculations: Thus we can write a ratio of the negative lift $F_{L,90}$ at $v = 90$ m/s to our result for the negative lift F_L at $v = 28.6$ m/s as

$$\frac{F_{L,90}}{F_L} = \frac{(90 \text{ m/s})^2}{(28.6 \text{ m/s})^2}.$$

Substituting our known negative lift of $F_L = 663.7$ N and solving for $F_{L,90}$ give us

$$F_{L,90} = 6572 \text{ N} \approx 6600 \text{ N}. \quad (\text{Answer})$$

Upside-down racing: The gravitational force is, of course, the force to beat if there is a chance of racing upside down:

$$\begin{aligned} F_g &= mg = (600 \text{ kg})(9.8 \text{ m/s}^2) \\ &= 5880 \text{ N}. \end{aligned}$$

With the car upside down, the negative lift is an *upward* force of 6600 N, which exceeds the downward 5880 N. Thus, the car could run on a long ceiling *provided* that it moves at about 90 m/s ($= 324 \text{ km/h} = 201 \text{ mi/h}$). However, moving that fast while right side up on a horizontal track is dangerous enough, so you are not likely to see upside-down racing except in the movies.

**QUÉ
MUESTRA**

Que en una curva plana la única fuerza horizontal es el rozamiento, así que es él el que tiene que valer $m v^2/R$. El libro no mezcla los ejes: el radial da $\mu_s F_N = m v^2/R$, el vertical da $F_N = mg + F_L$, y recién al combinarlos aparece $F_L = m (v^2/\mu_s R - g) = 663,7 \text{ N}$.

LA CLAVE

«Al borde de deslizar» es lo que habilita escribir $f_s = \mu_s F_N$. Es la misma condición del ejemplo anterior de fricción, pero usada al revés: allá se comparaba F_x contra $\mu_s F_N$ para *averiguar* si el bloque se movía; acá se sabe que está justo en el límite, y por eso la desigualdad se vuelve igualdad y se puede despejar.

SE PARECE A

P2 ej. 14 (el cuerpo que gira atado a un resorte: la centrípeta la da el resorte en vez del rozamiento, pero la ecuación radial se escribe igual) y **P2 ej. 18** (el cuerpo sobre el cono, donde también hay que separar radial de vertical antes de combinar). La lógica de $\mu_s F_N$ como cota es la de **P2 ej. 8 y 19**.

OJO CON

Otra vez la normal no es mg . La carga aerodinámica empuja hacia abajo y suma: $F_N = mg + F_L = 6543,7 \text{ N}$ contra los 5880 N del peso solo. Es el mismo detalle que el $F \sin \theta$ del bloque: cada fuerza vertical de más corre la normal, y con ella el umbral del rozamiento.